

PREPARED FOR

123 INC

SAMPLE

39.21470, -122.01400

T16N, R1W, Mount Diablo Meridian

✓ CORS Elev Delta: 97 ft above nearest CORS station (P270, 3.0 mi) – Good baseline geometry.

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
39.2147°N 122.0140°W WGS84	58 ft NAVD88 · MSL	CA 2 (0402) · N 2,204,196.94 E 6,557,700.15 (US ft, NAD83)	Zone X N/A	13.1° E WMM 2026

APN 001-073-001-000 | ZONING 0-S | ACRES 1.909 ac

BEST GNSS WINDOW Friday Jul 17 · 3:00 PM – 7:00 PM | PDOP 0.91 · 34 sv | NEAREST CORS
P270 · 3.0 mi ✓

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VIEW FIELD MAP

SITE SATELLITE

EMERGENCY MAP

SCAN FOR KMZ
Site file · Google Earth
Download KMZ

ASSESSOR-RECORDER
Colusa County
Clerk-Recorder
546 Jay St
Colusa, CA 95932
NEAREST BN
KT0441 · 0 mi
NEAREST CORS
P270 · 3.0 mi
GENERATED
July 17,
2026
SYSTEM
TopoScout ·
info@toposcout.com



INTERACTIVE REPORT

This report includes clickable links and QR codes throughout – tap or scan them to open maps, official data sources, and field tools directly. Your site is also available as a drafting-ready CAD (DXF) file with georeferenced aerials. Add it anytime at toposcout.com.

SITE & SIGNAL REPORT

COORDINATES

39.2147, -122.0140

WGS84 / NAD83

ELEVATION

58 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.1° E

Compass reads 13.1° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel N/A

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.1

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

16.9 TECU

-5.0 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.1). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Fri, Jul 17	0.99	3:00 PM – 7:00 PM	34	Ideal
Sat, Jul 18	0.98	1:00 PM – 5:00 PM	35	Ideal
Sun, Jul 19	0.98	3:00 PM – 7:00 PM	34	Ideal
Mon, Jul 20	0.99	3:00 PM – 7:00 PM	34	Ideal
Tue, Jul 21	0.98	3:00 PM – 7:00 PM	33	Ideal
Wed, Jul 22	1.00	4:00 PM – 8:00 PM	33	Good
Thu, Jul 23	1.00	7:00 AM – 11:00 AM	31	Good

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

Days analyzed: 7 Recommended work days: 5 of 7 Best PDOP: 0.98

►

RECOMMENDED WINDOW

Thu, Jul 23

6:00 AM – 10:00 AM

Avg wind 2 mph · Gusts 3 mph

Ideal conditions — 4-hr static occupation

ALTERNATE WINDOW

Sat, Jul 18

12:00 PM – 4:00 PM

Avg wind 5 mph · Gusts 8 mph

Ideal conditions — 4-hr static occupation

🔥

WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 17, 2026) · Conditions may change – verify before fieldwork

📶

GPS CONSTELLATION HEALTH

⚠️ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

MULTIPATH RISK

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–1.0 across survey windows

PDOP remains ≤ 1.0 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 95°F

Low 58°F

Wind max 14 mph

Gusts 22 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:32 PM

Waxing Crescent 8%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.91, 34 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	1.04		30 sv		GOOD
7:00 AM	1.12		27 sv		GOOD
8:00 AM	0.91		35 sv		IDEAL
9:00 AM	0.96		32 sv		IDEAL
10:00 AM	0.93		31 sv		IDEAL
11:00 AM	0.94		29 sv		IDEAL
12:00 PM	0.95		30 sv		IDEAL
1:00 PM	1.13		27 sv		GOOD
2:00 PM	1.03		31 sv		GOOD
3:00 PM	0.91		35 sv		IDEAL
4:00 PM	0.92		35 sv		IDEAL
5:00 PM	0.91		34 sv		IDEAL
6:00 PM	0.91		31 sv		IDEAL
7:00 PM	1.26		24 sv		GOOD



GO – GOOD CONDITIONS

Clear sky | Wind: 14 mph (gusts 21.5 mph) | Precip: 0" | Temp: 57.9°–94.9°F



DRONE WIND ANALYSIS



CAUTION Peak: 14/21.5 mph SSE

* Sunrise: 5:56 AM ☾ Sunset: 8:32 PM fly window clipped to daylight

✓ Best flight window: 6:00 AM – 7:00 PM (13 hrs)

🔄 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≠turbulence	DIR
5 AM	3/13 mph		SE
6 AM	4/10 mph		SE
7 AM	2/2 mph		N
8 AM	0/4 mph		W
9 AM	3/6 mph		NNW
10 AM	2/2 mph		NNW
11 AM	1/3 mph		SE
12 PM	5/7 mph		S
1 PM	8/9 mph		S
2 PM	9/11 mph		S
3 PM	11/13 mph		SSE
4 PM	13/15 mph		SSE
5 PM	13/17 mph		SSE
6 PM	14/19 mph		SSE
7 PM	14/22 mph		SSE
8 PM	13/20 mph		SSE

BEST WINDOW

6:00 AM – 7:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ▶ FREE TO FLY

Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#2

Saturday, Jul 18

WEEKEND

Clear sky

13 mph

GO

High 103°F Low 55°F Wind max 13 mph Gusts 23 mph Precip 0.00" Sunrise 5:56 AM Sunset 8:31 PM

Waxing Crescent 15%

GNSS: Avg PDOP 0.98 · Best 4-hr window 1:00 PM – 5:00 PM (PDOP 0.93, 35 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	1.01	31 sv	GOOD
7:00 AM	1.02	32 sv	GOOD
8:00 AM	1.03	32 sv	GOOD
9:00 AM	0.91	33 sv	IDEAL
10:00 AM	0.98	31 sv	IDEAL
11:00 AM	1.06	28 sv	GOOD
12:00 PM	0.96	30 sv	IDEAL
1:00 PM	0.97	32 sv	IDEAL
2:00 PM	0.91	35 sv	IDEAL
3:00 PM	0.91	35 sv	IDEAL
4:00 PM	0.93	36 sv	IDEAL
5:00 PM	0.99	32 sv	IDEAL
6:00 PM	0.91	34 sv	IDEAL
7:00 PM	1.18	29 sv	GOOD



GO - GOOD CONDITIONS

Clear sky | Wind: 13 mph (gusts 22.6 mph) | Precip: 0" | Temp: 54.6°–102.8°F



FIELD SAFETY WARNINGS



HIGH HEAT 102.8°F — Monitor crew for heat stress



DRONE WIND ANALYSIS



CAUTION Peak: 13/22.6 mph SSE

* Sunrise: 5:56 AM Sunset: 8:31 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)



Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	2/12 mph		SSE
6 AM	5/11 mph		SE
7 AM	5/8 mph		SSE
8 AM	1/3 mph		E
9 AM	4/7 mph		N
10 AM	4/6 mph		N
11 AM	2/2 mph		NW
12 PM	2/2 mph		WSW
1 PM	4/5 mph		SSW
2 PM	6/6 mph		SSW
3 PM	7/8 mph		S
4 PM	8/9 mph		SSE
5 PM	10/10 mph		SSE
6 PM	11/11 mph		SSE
7 PM	12/12 mph		S
8 PM	9/15 mph		S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ▶ FREE TO FLY

Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#3

Sunday, Jul 19

Overcast

10 mph

GO

WEEKEND

High 104°F

Low 61°F

Wind max 10 mph

Gusts 16 mph

Precip 0.00"

Sunrise 5:57 AM

Sunset 8:31 PM

Waxing Crescent 23%

GNSS: Avg PDOP **0.98** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.94, 34 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	0.96		34 sv	IDEAL
7:00 AM	0.93		31 sv	IDEAL
8:00 AM	0.93		33 sv	IDEAL
9:00 AM	0.94		31 sv	IDEAL
10:00 AM	1.09		28 sv	GOOD
11:00 AM	1.14		25 sv	GOOD
12:00 PM	0.99		28 sv	IDEAL
1:00 PM	0.95		31 sv	IDEAL
2:00 PM	0.99		33 sv	IDEAL
3:00 PM	0.95		33 sv	IDEAL
4:00 PM	0.95		36 sv	IDEAL
5:00 PM	0.89		35 sv	IDEAL
6:00 PM	0.95		32 sv	IDEAL
7:00 PM	1.07		29 sv	GOOD

GO — GOOD CONDITIONS

Partly cloudy | Wind: 9.6 mph (gusts 16.1 mph) | Precip: 0" | Temp: 60.7°–103.5°F

FIELD SAFETY WARNINGS

HIGH HEAT 103.5°F — Monitor crew for heat stress

DRONE WIND ANALYSIS

GO Peak: 9.6/16.1 mph SSE

* Sunrise: 5:57 AM ☾ Sunset: 8:31 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	4/3 mph		ENE
6 AM	3/4 mph		NNE
7 AM	4/5 mph		N
8 AM	5/5 mph		N
9 AM	5/4 mph		NNW
10 AM	4/3 mph		NW
11 AM	2/2 mph		N
12 PM	2/3 mph		ESE
1 PM	3/3 mph		S
2 PM	5/5 mph		SSW
3 PM	5/6 mph		SSW
4 PM	4/7 mph		S
5 PM	6/8 mph		SSE
6 PM	10/9 mph		SSE
7 PM	10/15 mph		SSW
8 PM	8/16 mph		SSW

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ► FREE TO FLY
Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107
→ TFR STATUS NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 95°F

Low 72°F

Wind max 11 mph

Gusts 15 mph

Precip 0.00"

Sunrise 5:58 AM

Sunset 8:30 PM

Waxing Crescent 33%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.93, 34 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	0.91		34 sv		IDEAL
7:00 AM	0.92		35 sv		IDEAL
8:00 AM	0.93		35 sv		IDEAL
9:00 AM	1.00		30 sv		GOOD
10:00 AM	0.98		28 sv		IDEAL
11:00 AM	1.23		24 sv		GOOD
12:00 PM	0.93		29 sv		IDEAL
1:00 PM	1.08		26 sv		GOOD
2:00 PM	1.02		32 sv		GOOD
3:00 PM	0.98		34 sv		IDEAL
4:00 PM	0.94		33 sv		IDEAL
5:00 PM	0.92		33 sv		IDEAL
6:00 PM	0.87		35 sv		IDEAL
7:00 PM	1.19		28 sv		GOOD

GO – GOOD CONDITIONS

Partly cloudy | Wind: 10.8 mph (gusts 15.2 mph) | Precip: 0" | Temp: 71.5°–95°F

DRONE WIND ANALYSIS ✓ GO Peak: 10.8/15.2 mph NE

* Sunrise: 5:58 AM ☾ Sunset: 8:30 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	3/4 mph		SSE
6 AM	2/3 mph		ESE
7 AM	4/3 mph		E
8 AM	1/2 mph		S
9 AM	3/4 mph		SW
10 AM	1/4 mph		W
11 AM	3/5 mph		NW
12 PM	4/7 mph		WNW
1 PM	4/6 mph		WNW
2 PM	5/7 mph		WNW
3 PM	6/8 mph		NW
4 PM	7/9 mph		NNW
5 PM	7/8 mph		NW
6 PM	5/6 mph		NNW
7 PM	4/4 mph		WNW
8 PM	5/5 mph		NW

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ▶ FREE TO FLY

Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 105°F

Low 74°F

Wind max 13 mph

Gusts 20 mph

Precip 0.00"

Sunrise 5:59 AM

Sunset 8:29 PM

Waxing Crescent 43%

GNSS: Avg PDOP **0.98** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.94, 33 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	0.91	33 sv	IDEAL
7:00 AM	0.89	36 sv	IDEAL
8:00 AM	0.93	33 sv	IDEAL
9:00 AM	1.15	29 sv	GOOD
10:00 AM	1.01	31 sv	GOOD
11:00 AM	1.07	28 sv	GOOD
12:00 PM	0.99	29 sv	IDEAL
1:00 PM	1.06	28 sv	GOOD
2:00 PM	0.98	31 sv	IDEAL
3:00 PM	0.87	35 sv	IDEAL
4:00 PM	0.94	36 sv	IDEAL
5:00 PM	1.00	31 sv	GOOD
6:00 PM	0.96	30 sv	IDEAL
7:00 PM	1.02	29 sv	GOOD

GO — GOOD CONDITIONS
Partly cloudy | Wind: 13.2 mph (gusts 19.5 mph) | Precip: 0" | Temp: 74.2°–105°F

FIELD SAFETY WARNINGS
 HIGH HEAT 105°F — Monitor crew for heat stress

DRONE WIND ANALYSIS

☒ **GO** Peak: 13.2/19.5 mph SSW

* Sunrise: 5:59 AM ☾ Sunset: 8:29 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	1/2 mph		NE
6 AM	2/2 mph		SSW
7 AM	3/3 mph		SSE
8 AM	3/4 mph		SSW
9 AM	3/1 mph		ESE
10 AM	6/5 mph		E
11 AM	5/5 mph		ESE
12 PM	6/6 mph		ESE
1 PM	5/7 mph		SE
2 PM	5/8 mph		SSE
3 PM	6/9 mph		SSW
4 PM	9/9 mph		SW
5 PM	12/10 mph		WSW
6 PM	13/13 mph		WSW
7 PM	13/15 mph		SW
8 PM	12/17 mph		SW

BEST WINDOW
6:00 AM – 9:00 PM
Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS G** ▶ **FREE TO FLY**
Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ **NO TFRs WITHIN 30 MI** [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 104°F

Low 71°F

Wind max 17 mph

Gusts 19 mph

Precip 0.00"

Sunrise 6:00 AM

Sunset 8:29 PM

First Quarter 53%

GNSS: Avg PDOP **1.00** · Best 4-hr window 4:00 PM – 8:00 PM (PDOP 0.93, 33 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	1.09		30 sv	GOOD
7:00 AM	1.05		30 sv	GOOD
8:00 AM	0.88		37 sv	IDEAL
9:00 AM	1.00		31 sv	GOOD
10:00 AM	1.02		29 sv	GOOD
11:00 AM	1.17		25 sv	GOOD
12:00 PM	0.97		28 sv	IDEAL
1:00 PM	1.05		28 sv	GOOD
2:00 PM	0.98		34 sv	IDEAL
3:00 PM	1.08		33 sv	GOOD
4:00 PM	0.90		35 sv	IDEAL
5:00 PM	0.93		33 sv	IDEAL
6:00 PM	0.96		31 sv	IDEAL
7:00 PM	0.94		32 sv	IDEAL



GO — GOOD CONDITIONS

Clear sky | Wind: 16.5 mph (gusts 19 mph) | Precip: 0" | Temp: 70.9°–103.5°F



FIELD SAFETY WARNINGS



HIGH HEAT 103.5°F — Monitor crew for heat stress



DRONE WIND ANALYSIS



CAUTION Peak: 16.5/19 mph SSE

* Sunrise: 6:00 AM Sunset: 8:29 PM fly window clipped to daylight

✓ Best flight window: 6:00 AM – 5:00 PM (11 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	8/13 mph		SE
6 AM	8/11 mph		SE
7 AM	8/9 mph		SE
8 AM	8/8 mph		SE
9 AM	8/7 mph		SE
10 AM	7/7 mph		SE
11 AM	7/7 mph		SE
12 PM	7/9 mph		SE
1 PM	9/11 mph		SSE
2 PM	11/13 mph		SSE
3 PM	12/14 mph		S
4 PM	15/16 mph		SSW
5 PM	17/17 mph		SSW
6 PM	16/18 mph		SSW
7 PM	13/19 mph		SSW
8 PM	11/19 mph		SSW

BEST WINDOW

6:00 AM – 5:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ► FREE TO FLY

Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 108°F

Low 66°F

Wind max 15 mph

Gusts 24 mph

Precip 0.00"

Sunrise 6:00 AM

Sunset 8:28 PM

First Quarter 64%

GNSS: Avg PDOP **1.00** · Best 4-hr window 7:00 AM – 11:00 AM (PDOP 0.97, 31 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	1.03		30 sv	GOOD
7:00 AM	0.93		34 sv	IDEAL
8:00 AM	0.94		32 sv	IDEAL
9:00 AM	1.08		27 sv	GOOD
10:00 AM	0.92		31 sv	IDEAL
11:00 AM	1.09		26 sv	GOOD
12:00 PM	0.98		31 sv	IDEAL
1:00 PM	0.99		30 sv	IDEAL
2:00 PM	0.97		31 sv	IDEAL
3:00 PM	1.00		33 sv	GOOD
4:00 PM	1.00		32 sv	GOOD
5:00 PM	0.98		33 sv	IDEAL
6:00 PM	0.95		31 sv	IDEAL
7:00 PM	1.10		26 sv	GOOD



GO – GOOD CONDITIONS

Clear sky | Wind: 14.8 mph (gusts 23.9 mph) | Precip: 0" | Temp: 66.4°–107.5°F



FIELD SAFETY WARNINGS

 EXTREME HEAT 107.5°F — Heat safety protocols required



DRONE WIND ANALYSIS

 CAUTION Peak: 14.8/23.9 mph S

* Sunrise: 6:00 AM Sunset: 8:28 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 7:00 PM (13 hrs)

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	2/2 mph		E
6 AM	2/2 mph		NE
7 AM	2/3 mph		N
8 AM	3/3 mph		NNW
9 AM	3/2 mph		NW
10 AM	2/1 mph		NW
11 AM	1/1 mph		WNW
12 PM	2/4 mph		SSW
1 PM	4/7 mph		S
2 PM	7/10 mph		S
3 PM	9/12 mph		S
4 PM	11/12 mph		SSE
5 PM	12/14 mph		SSE
6 PM	11/17 mph		S
7 PM	13/22 mph		SW
8 PM	15/24 mph		SW

BEST WINDOW

6:00 AM – 7:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ► FREE TO FLY
Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

GNSS SURVEY — COLUSA COUNTY, CA

Friday, July 17, 2026

=====

TODAY (FRIDAY, JULY 17, 2026)

STATUS: GO

RATIONALE:

Clear skies, excellent PDOP (0.91–0.99 avg), 34 satellites available, moderate ionospheric activity manageable with standard occupation times. Wind 14 mph is acceptable. Best window 3:00 PM–7:00 PM (PDOP 0.91). PRN 26 maintenance window (00:35–12:35 UTC) does not impact afternoon sessions. No weather impediments.

ACTION: Crew may mobilize and conduct static observations this afternoon onward.

=====

MONDAY CALL (JULY 20 — CREW MOBILIZATION DECISION)

STATUS: GO

RATIONALE:

Monday is a regular working day with PDOP 0.93–0.99 (avg 0.99), 34 satellites, clear skies improving into the 3:00 PM–7:00 PM optimal window. Ionospheric activity remains moderate. No satellite health impacts. Conditions are suitable for full-day field operations.

ACTION: Confirm crew deployment for Monday start. Plan primary static sessions 3:00 PM–7:00 PM.

=====

WEEK RANKING (BEST TO WORST FOR GNSS SURVEY)

RANK 1: FRIDAY, JULY 17 (TODAY)

Rationale: PDOP 0.91 (best of week), clear sky, 34 sats, no weather barriers. Afternoon window 3–7 PM is optimal.

RANK 2: MONDAY, JULY 20

Rationale: PDOP 0.93, 34 sats, clear sky, excellent afternoon window 3–7 PM. Best working weekday.

RANK 3: WEDNESDAY, JULY 22

Rationale: PDOP 0.93, clear sky, 33 sats. Evening window 4–8 PM slightly later but still strong. Wind higher (17 mph) but manageable.

RANK 4: SATURDAY, JULY 18 (WEEKEND)

Rationale: PDOP 0.93, 35 sats (highest constellation), clear sky, best noon–5 PM window. Weekend day but conditions are superior. High heat (103 °F) — crew fatigue factor.

RANK 5: TUESDAY, JULY 21

Rationale: PDOP 0.94, 33 sats, overcast (still acceptable), 3–7 PM window strong. Overcast introduces slight ionospheric uncertainty; rank lower than clear days.

RANK 6: SUNDAY, JULY 19 (WEEKEND)

Rationale: PDOP 0.94, 34 sats, but OVERCAST conditions degrade visibility and introduce ionospheric delay risk. Weekend. Best window 3–7 PM still viable but caution advised.

RANK 7: THURSDAY, JULY 23

Rationale: PDOP 0.97 (worst), only 31 sats, morning-only best window (7–11 AM, PDOP 0.97). Lowest constellation availability. Highest

heat (108 °F). Recommend avoid or use as backup only.

=====

BEST STATIC WINDOW (4-HOUR CONSECUTIVE SESSION)

RECOMMENDED WINDOW: FRIDAY, JULY 17, 3:00 PM–7:00 PM

Secondary option: MONDAY, JULY 20, 3:00 PM–7:00 PM

Rationale: Friday 3–7 PM delivers PDOP 0.91, 34 satellites, clear sky, and mild thermal conditions (cooling toward 58 °F overnight). This is the single strongest geometry window of the week. Ionospheric moderate activity does not degrade 4-hour static significantly. If Friday cannot be executed, Monday 3–7 PM is statistically equivalent (PDOP 0.93) and offers full crew availability post-mobilization.

=====

DRONE ASSESSMENT (VFR + GNSS-dependent operations)

FRIDAY, JULY 17: FLYABLE

Clear, 14 mph wind (acceptable), good PDOP 0.91. Optimal for RTK/PPK workflows.

SATURDAY, JULY 18: CAUTION

Clear, but extreme heat (103 °F) increases thermal drift risk on airframe and battery. Maintain conservative flight time (15 min max). Weekend.

SUNDAY, JULY 19: CAUTION

Overcast reduces visual orientation. PDOP 0.94 adequate but ionospheric delay (TEC -5.0 TECU) may degrade RTK baseline. Flight operations possible but use conservative planning.

MONDAY, JULY 20: FLYABLE

Overcast clearing by afternoon, PDOP 0.93, acceptable wind 11 mph. RTK performance good 3–7 PM.

TUESDAY, JULY 21: CAUTION

Overcast, high heat (105 °F), 13 mph wind. PDOP 0.94 acceptable but visibility limited. Evening session 3–7 PM preferred.

WEDNESDAY, JULY 22: FLYABLE

Clear, PDOP 0.93, but elevated wind (17 mph). Acceptable for experienced crews. Avoid low-altitude hover ops. Evening window 4–8 PM best.

THURSDAY, JULY 23: NO-FLY

Worst PDOP (0.97), only 31 sats, extreme heat (108 °F), 15 mph wind. Highest risk for RTK loss of lock and airframe thermal shutdown. Defer if possible.

=====

FIELD WARNINGS

1. EXTREME HEAT ADVISORY (WEEKEND + TUESDAY)

Temperatures 103–105 °F forecast Saturday, Sunday, Tuesday. Crew hydration mandatory. Battery cycling (GNSS receivers, drones, radios) will degrade. Plan shade/cooling stations. Personnel fatigue increases error risk. Limit midday operations; shift to early morning or evening windows.

2. IONOSPHERIC DEGRADATION — MONITOR

TEC anomaly -5.0 TECU (below 30-day median) flagged with non-zero quality flag. Estimate reliability reduced. Static sessions should extend occupation to 6+ hours (vs. standard 4 hours) to compensate. Real-time RTK users: expect occasional baseline instability; validate with post-processing.

3. SATELLITE HEALTH

PRN 11 remains unhealthy (UNUSUFN); no recovery ETA. Receivers will auto-exclude. Net impact minimal given 34-satellite availability.

PRN 26 maintenance (Fri 00:35–12:35 UTC) is overnight UTC; confirm local time impact with receiver firmware. PRN 3 maintenance occurred before reporting window — informational only.

4. WIND HAZARD (WEDNESDAY)

17 mph gusts Wednesday 4–8 PM. Secure all loose equipment and antenna setup. Drone operations at elevated risk.

5. NO WILDFIRE THREAT

NIFC reports no active fires in region. Air quality unrestricted.

=====

OVERALL RECOMMENDATION

Execute primary static campaign FRIDAY AFTERNOON (3–7 PM) with 4-hour occupation minimum. Mobilize full crew Monday for secondary sessions (Mon 3–7 PM, Wed 4–8 PM, Sat morning as backup). Avoid Thursday entirely and defer Sunday to lower-ranking contingency. Extend all static occupation times to 6 hours due to ionospheric TEC anomaly. Enforce heat mitigation protocols for weekend operations. Drone RTK should prioritize Friday, Monday, and Wednesday windows; avoid Thursday and Sunday.

=====

Report prepared: Friday, 17 July 2026

Analyst: GNSS Operations Planning

Valid through: Friday, 24 July 2026

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Colusa County Clerk-Recorder

546 Jay St, Colusa, CA 95932

ASSESSOR / TAX OFFICE

Colusa County Assessor

546 Jay St, Colusa, CA 95932

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P270	HOPKINSLGHCN2005	3.0 mi
 NGS ↗	SUTB	SUTTER BUTTES	10.4 mi
 NGS ↗	CH01	CHICO 1	24.0 mi
 NGS ↗	CH05	CHICO 5	24.0 mi
 NGS ↗	CH02	CHICO 2	24.0 mi
 NGS ↗	CH06	CHICO 6	24.0 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	KT0441	"58 B"	18.660m / 61.2ft NAVD88	Order	0 mi
 Datasheet ↗	KT2287	"60 B RESET"	19.300m / 63.3ft NAVD88	Order	0.3 mi
 Datasheet ↗	KT0440	"W"	18.180m / 59.6ft NAVD88	Order	0.3 mi
 Datasheet ↗	KT0443	"X"	18.480m / 60.6ft NAVD88	Order	0.7 mi
 Datasheet ↗	KT1843	"GREEN"	15.400m / 50.5ft NAVD88	Order	1.9 mi
 Datasheet ↗	KT1844	"TROUGH"	15.810m / 51.9ft NAVD88	Order	2.3 mi
 Datasheet ↗	DL9151	"COLIND"	14.850m / 48.7ft NAVD88	Order	2.3 mi
 Datasheet ↗	KS1856	"COLUSA"	17.000m / 55.8ft NAVD88	Order	2.3 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: N/A

Effective: N/A

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
KT0441	58 B	39.215278	-122.013611	61.2	
KT2287	60 B RESET	39.213889	-122.008333	63.3	
KT0440	W	39.210361	-122.016250	59.6	
KT0443	X	39.212222	-122.001111	60.6	
KT1843	GREEN	39.188010	-122.023061	50.5	
KT1844	TROUGH	39.195512	-122.048830	51.9	
DL9151	COLIND	39.185125	-121.994591	48.7	
KS1856	COLUSA	39.192905	-121.981688	55.8	

[Download .txt File](#)

USGS STREAM GAUGES – within 25 miles

STATION	DIST	LEVEL	FLOW	FLOOD STATUS
SACRAMENTO R A COLUSA CA #11389500	0.7 mi	43.94 ft	8,610 cfs	● Normal
SACRAMENTO R BL WILKINS SLOUGH NR GRIMES CA #11390500	17.4 mi	31.03 ft	7,110 cfs	● Normal

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	49.7 mi	4.84 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Oroville	36.1 mi	856.21 ft (-44 ft)	900 ft	2,791K af	79%
Black Butte Lake	44.5 mi	—	228 ft	71K af	53%
Folsom Lake	57.5 mi	445.4 ft (-21 ft)	466 ft	759K af	78%
Shasta Lake	106.1 mi	1,021.39 ft (-46 ft)	1,067 ft	3,321K af	73%
Lake Del Valle	112.6 mi	696.51 ft (-9 ft)	706 ft	35K af	67%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Whiskeytown Lake	100.4 mi	1,209 ft*	1,209 ft	238K af	99%
Trinity Lake	114.6 mi	2,370 ft*	2,370 ft	1,948K af	80%
New Melones Lake	118.2 mi	1,088 ft*	1,088 ft	1,655K af	69%

ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Black Butte-Pool	47 mi	456.2 ft	458.36 ft	—	100%
Englebright Lake-Pool	52 mi	524.1 ft	522.35 ft	—	100%
Camanche	97 mi	230 ft	221.86 ft	—	104%
Del Valle	112 mi	696.5 ft	695.01 ft	—	100%
Farmington Dam-Pool	117 mi	123.8 ft	117.09 ft	—	106%
Boca	133 mi	5,601.3 ft	5,579.88 ft	—	100%

SNOTEL STATIONS (NRCS) – within 50 miles

STATION	DIST	ELEV	SWE	DEPTH	TEMP	ACCESS
Noel Spring [Snow Course]	41.4 mi	4,950 ft	0"	—	—	● Clear

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
014N001W28E001M	12.5 mi	364 ft	—	—	S100CNRLVL
014N001W30N002M	12.7 mi	394 ft	35.00 ft BGS	1960-02-23	S100CNRLVL
014N001W30N001M	12.9 mi	180 ft	44.00 ft BGS	1960-04-01	S100CNRLVL
014N001W32E002M	13.3 mi	508 ft	31.00 ft BGS	1966-10-14	S100CNRLVL
014N002W29J001M	13.3 mi	412 ft	—	—	S100CNRLVL

015N002E08R002M	13.8 mi	70 ft	35.00 ft BGS	1968-02-04	S100CNRLVL
014N002W34J001M	13.8 mi	463 ft	—	—	S100CNRLVL
014N001W32P001M	13.9 mi	100 ft	—	—	S100CNRLVL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Central Valley aquifer system (S100CNRLVL) consists of alluvial and lacustrine deposits laid down over millions of years as the Sacramento River and its tributaries built up thick sequences of sand, silt, and clay across the valley floor. These sediments were deposited in a structural basin and are highly productive where sandy and gravelly layers predominate. At this Colusa County location, water availability depends on whether the well penetrates the coarser, more permeable zones within this layered system.

FORMATION

Name	Central Valley undifferentiated (alluvial and lacustrine deposits)
Age / Rock	Quaternary alluvium and Tertiary lacustrine deposits; predominantly unconsolidated sand, silt, clay, and gravel.
Extent	est. 16,000–22,000 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	70–508 ft (7–51 floors)
Typical Depth	200–350 ft (20–35 floors)
Recharge Source	Precipitation on northern Sierra Nevada and Cascade uplands; lateral inflow from bedrock highlands; canal seepage and return flow from agricultural irrigation
Transit Time	15–50 years to 300 ft depth (est. USGS/DWR)

WATER QUALITY

pH	est. 6.8–7.5 (est. USGS/DWR)
Hardness	est. 150–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 250–500 mg/L (est. USGS/DWR)
Notes	No sample data available. Central Valley agriculture may contribute nitrate; typical for formation in actively farmed areas.

TREND

Trend	Water level declined from ~31 ft BGS (1966) to 35–44 ft BGS (1960, 1968), indicating multi-decadal drawdown consistent with regional agricultural pumping.
Source	On-site historical measurements (1960–1968)
Seasonal Swing	est. 8–12 ft seasonal swing (est. USGS/DWR regional data)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The site at 39.2147, -122.0140 is located in the North Coast region of California where volcanic and consolidated bedrock geology predominates, providing natural confinement and structural stability to underlying aquifer systems. The Central Valley aquifer code S100CNRLVL represents a consolidated, relatively low-yield formation that does not exhibit the thick, compressible clay sequences characteristic of subsidence-prone basins. Because this area is not within a critically overdrafted groundwater basin and maintains moderate drainage conditions, groundwater withdrawal has not induced significant aquifer compaction, making this location geologically stable for survey monumentation and elevation referencing.

SUBSIDENCE ASSESSMENT

Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin

BENCHMARK IMPLICATIONS

Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate

SURVEY ACTION

Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	93 mph
Ground Snow Load	6.7 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	6 in minimum

Source: ASCE 7-22 Hazard Tool · gis.asce.org

CLIMATE ZONE (IECC 2021): Zone 3B · Hot-Dry · Hot-Dry - warm/hot summers, cool winters, dry (Central CA, NV, AZ) County: Colusa

AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This hot-dry climate demands a well-insulated, tight envelope with minimal air leakage to manage intense solar gain and temperature swings between hot days and cool nights. Concentrate glazing on north-facing elevations while minimizing or shading south and west-facing windows; if east-facing glass is necessary, use deep overhangs or exterior shading to block morning sun. The most effective passive strategy for this zone is thermal mass paired with night ventilation—design the home to absorb daytime heat in concrete floors or masonry walls, then open windows after sunset to allow the cool night air to purge that stored heat and pre-cool the building for the next day.

AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This Central California site in a hot-dry climate presents moderate wind exposure and light snow loads, with anticipated conventional spread footing conditions on moderately well-drained silt loam soils. Seismic parameters are unavailable (null values); a geotechnical investigation and seismic hazard assessment may warrant early confirmation to establish design criteria. Shallow frost depth and drainage characteristics suggest standard foundation detailing, pending site-specific soil confirmation.

FOUNDATION

Type Conventional spread footing anticipated
Basis Moonbend silt loam (Site Class D, moderately well-drained, Hydro Group C) suggests bearing capacity likely adequate for typical structures; however, consult a licensed engineer to confirm bearing capacity, settlement, and expansive potential before final design

STRUCTURAL IMPLICATIONS

Seismic Seismic design category and spectral accelerations unavailable (null values) — geotechnical investigation and USGS hazard assessment recommended to establish SDC and confirm whether ordinary or special seismic detailing may be warranted
Wind Exposure Exposure C (open terrain) anticipated per ASCE 7-22 Risk Category II; verify site obstructions and topography match exposure classification
Wind 93 mph design wind speed (3-sec gust) anticipated — standard ASCE 7-22 wind load procedures likely applicable; no hurricane or coastal high-wind exposure anticipated
Snow Light ground snow load of 6.7 psf anticipated — standard roof design likely adequate; roof configuration and thermal resistance should reflect IECC Zone 3B requirements

INVESTIGATIONS & INSPECTIONS

Geotech Required Yes — geotechnical investigation recommended to confirm Site Class D assignment, bearing capacity, liquefaction screening (if applicable), and settlement potential
Liquefaction Screen Screening recommended — Site Class D with moderately well-drained soils suggests lower risk, but confirm groundwater depth and soil composition with geotechnical study
Special Inspections Not anticipated at this stage pending SDC confirmation; consult engineer once seismic design category is established
Code Edition ASCE 7-22 / IBC 2021 (planning reference only)
Site Class Note Site Class D assumed for planning — geotechnical investigation with standard penetration testing or cone penetration testing recommended to confirm classification and soil parameters

SOILS — USDA SOIL SURVEY

MAP UNIT: Moonbend silt loam, 0 to 2 percent slopes **SERIES:** Moonbend
DRAINAGE CLASS: Moderately well drained **HYDRO GROUP:** C
SOIL ORDER: Mollisols **SUBORDER:** Xerolls

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Moonbend silt loam is a moderately well-drained Mollisol (Xeroll suborder) formed in alluvial or lacustrine silts and clays on nearly level valley floors and low terraces. The dark surface layer reflects accumulated organic matter typical of grassland-derived soils in semi-arid climates. A surveyor or developer should expect moderate bearing capacity, seasonal water table fluctuations within 3–4 feet, and careful moisture management during grading and fill placement to avoid subgrade softening.

Parent material: alluvial or lacustrine silt and clay; landscape position: valley floor and low terrace; vegetation: native grassland.

DRAINAGE

Class Moderately well drained
Seasonal Saturation 24–36 inches (spring/winter in wet years)
Flooding Risk Minimal on 0–2% slopes; assess local drainage patterns and stream proximity
Ponding Risk Low to none on well-maintained slopes

ENGINEERING

Bearing Capacity 2,500–3,500 psf (shallow foundations, assuming firm subgrade)
Rating Fair
Shrink-Swell Moderate — silt loam to clay loam substratum; monitor during moisture fluctuations
Frost Action Moderate — silt loam susceptible to frost heave in cold climates
Cut Suitability Good — material is generally cohesive and stable when exposed
Fill Suitability Fair — requires moisture control and compaction verification; clay substratum prone to rutting when wet

SEPTIC

Suitability Conditional
Notes Perc test required — seasonal saturation and clay-rich substratum may limit infiltration and effluent dispersal. High water table risk in wet years.

HAZARDS

Excavation Low to moderate — cohesive silt and clay; manageable with standard equipment
Expansive Clay Low — typical Mollisol clay mineralogy; monitor if subsoil clay content exceeds 30%
Erosion Low on maintained slopes; moderate risk in disturbed areas and concentrated flow paths

Data sources: USGS National Water Information System · waterdata.usgs.gov · California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · NRCS SNOTEL/AWDB · wcc.sc.egov.usda.gov · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 17, 2026, 2:34 AM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

200 total **1** active **180** plugged By status: **180** Plugged **11** Canceled **8** Idle **1** Active

By type: 153 DH 40 DG 4 OG 3 GAS

Lease / Well	Status	Type	Operator	Depth	Dist
TXO Production Corp.	Plugged	DH	TXO Production Corp.	—	1.2 mi
Great Basins Petroleum Co.	Plugged	DH	Great Basins Petroleum Co.	—	1.3 mi
Venoco, LLC	Plugged	DH	Venoco, LLC	—	1.4 mi
Phillips Petroleum Company	Plugged	DH	Phillips Petroleum Company	—	1.7 mi
Whiting Oil and Gas Corporation	Plugged	DG	Whiting Oil and Gas Corporation	—	2.3 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	2.5 mi
Northern Michigan Exploration Company	Plugged	DH	Northern Michigan Exploration C...	—	2.6 mi
Union Oil Company of California	Plugged	DH	Union Oil Company of California	—	2.6 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	2.7 mi
Whiting Oil and Gas Corporation	Plugged	DG	Whiting Oil and Gas Corporation	—	2.7 mi
Western Continental Operating Company	Plugged	DG	Western Continental Operating ...	—	2.7 mi
Great Basins Petroleum Co.	Plugged	DH	Great Basins Petroleum Co.	—	2.7 mi
Natural Gas Corp. of Calif.	Plugged	DH	Natural Gas Corp. of Calif.	—	2.8 mi
Exxon Mobil Corporation	Plugged	DH	Exxon Mobil Corporation	—	2.8 mi
Natural Gas Corp. of Calif.	Plugged	DH	Natural Gas Corp. of Calif.	—	2.8 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	2.8 mi
California Resources Production Corporation	Idle	DG	California Resources Production...	—	3 mi
Richard S. Rheem, Operator	Plugged	DH	Richard S. Rheem, Operator	—	3 mi
Key Production Company, Inc.	Plugged	DG	Key Production Company, Inc.	—	3 mi
Whiting Oil and Gas Corporation	Plugged	DG	Whiting Oil and Gas Corporation	—	3 mi
Natural Gas Corp. of Calif.	Plugged	DH	Natural Gas Corp. of Calif.	—	3.1 mi
California Resources Production Corporation	Idle	DG	California Resources Production...	—	3.3 mi
Jacobson-Colusa Oil Co.	Plugged	DH	Jacobson-Colusa Oil Co.	—	3.3 mi
Amerada Hess Corp.	Plugged	DH	Amerada Hess Corp.	—	3.4 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	3.4 mi
Crest Oil & Gas Management Corp.	Plugged	DG	Crest Oil & Gas Management C...	—	3.5 mi
Crest Oil & Gas Management Corp.	Plugged	DG	Crest Oil & Gas Management C...	—	3.5 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	3.5 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	3.6 mi
Pueblo Oil & Gas, Inc.	Plugged	DG	Pueblo Oil & Gas, Inc.	—	3.6 mi

Showing 30 of 200 wells. Full dataset via California.

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well, a plugged TXO Production Corp. well 1.2 miles away, is beyond all actionable review distances. With only 1 active producer within the 10-mile radius and no injection wells present, standard surface rights and easement due diligence during title review will be sufficient.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200
1
39

total in search area
within 5 mi
within 25 mi

Site Name	Status	Commodities	Distance
Cortina Mill	Producer	Sand/Gravel	0.6 mi
Lucky Boy Occurrences 2	Occurrence	Chromium	8.6 mi
Unnamed Prospects	Occurrence	Chromium	8.7 mi
Quarry	Unknown	STN	9.6 mi
Number Two	Unknown	Gold	10.1 mi
Secret Town	Unknown	Gold	10.2 mi
John Nugent Placer	Unknown	Gold	10.3 mi
San Juan Mine	Plant	Gold	10.4 mi
Myers Quarry	Producer	Stone/Crushed	10.7 mi
North Fork American River	Unknown	Gold	10.9 mi
Sutter Buttes Quarry	Producer	Sand/Gravel	11.4 mi
Sunset & Riverside	Occurrence	Gold	11.5 mi
Butte Ranch Borrow Site	Producer	Sand/Gravel	11.7 mi
Morehead Pit & Mill	Producer	Sand/Gravel	11.9 mi
Lower Sand Creek Bar	Producer	Sand/Gravel	12.7 mi
Pennsylvania	Unknown	Gold	12.9 mi
Butte Rock North Pit	Producer	Stone/Crushed	14.1 mi
Sand Creek Pit	Producer	Sand/Gravel	14.7 mi
Mcgilvray Quarry	Past Producer	Sand/Gravel	17.6 mi
Golden Gate Consolidated Qtz. Mine	Past Producer	Gold	17.7 mi
Antelope Valley Deposit	Occurrence	CLY_BN	18.6 mi
Salt Creek Bar	Producer	Sand/Gravel	19.1 mi
Robinson Construction Co	Producer	Sand/Gravel	19.6 mi
Mcfarland Pit and Mill	Producer	Sand/Gravel	19.8 mi
Gravel Pits	Unknown	Sand/Gravel	21.8 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

The Cortina Mill (0.6 miles away) is an active sand and gravel producer and will be the primary mineral resource consideration for this site. Expect heavy truck traffic on local access roads during operating hours, along with potential dust generation from aggregate handling and processing operations. No other active producers are located within the 5-mile radius.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

2
2
4

Endangered
Threatened
Total w/ Critical Habitat

2 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
Butte County meadowfoam	<i>Limnanthes floccosa ssp. californica</i>	Endangered

COMMON NAME	SCIENTIFIC NAME	STATUS
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
California tiger Salamander	<i>Ambystoma californiense</i>	Threatened
Yellow-billed Cuckoo	<i>Coccyzus americanus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Review – 39.2147, -122.0140 Section 7 consultation with USFWS will be triggered if the proposed project involves federal permits, funding, or approvals, or if it may affect critical habitat or listed species through direct take, habitat modification, or hydrological changes. The California tiger salamander and vernal pool tadpole shrimp present the most significant planning constraints due to their dependence on seasonal wetlands and the site's location within the Central Valley vernal pool ecosystem. Butte County meadowfoam and yellow-billed cuckoo habitat requirements should also be evaluated relative to site hydrology and riparian conditions. The recommended next step is to conduct a focused biological assessment and wetland delineation to map critical habitat boundaries, establish baseline conditions, and determine project-specific avoidance and minimization measures before finalizing site design and permitting strategy. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

53	222	2026-07-07
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
alkali heliotrope	<i>Heliotropium curassavicum</i>	2025-08-15
alkali mallow	<i>Malvella leprosa</i>	2024-07-29
Alkali Weed	<i>Cressa truxillensis</i>	2024-05-15
big saltbush	<i>Atriplex lentiformis</i>	2024-01-16
Bonfire moss	<i>Funaria hygrometrica</i>	2023-11-30
box elder	<i>Acer negundo</i>	2025-05-21
broadleaf mistletoe	<i>Phoradendron macrophyllum</i>	2024-12-18
California brome	<i>Bromus carinatus</i>	2024-04-12
California mugwort	<i>Artemisia douglasiana</i>	2024-04-27
California wild grape	<i>Vitis californica</i>	2026-06-13
California Wild Rose	<i>Rosa californica</i>	2025-12-10
Common Madia	<i>Madia elegans</i>	2023-07-06
common tule	<i>Schoenoplectus acutus occidentalis</i>	2024-05-15
coyote brush	<i>Baccharis pilularis</i>	2023-09-06
cutleaf groundcherry	<i>Physalis angulata</i>	2025-07-15
Erect burhead	<i>Echinodorus berteroi</i>	2023-09-02
Frémont Cottonwood	<i>Populus fremontii</i>	2023-09-02
Giant Blazingstar	<i>Mentzelia laevicaulis</i>	2024-07-28
giant wild rye	<i>Leymus condensatus</i>	2026-03-02
heart-leaf milkweed	<i>Asclepias cordifolia</i>	2025-04-10
horseweed	<i>Erigeron canadensis</i>	2026-05-29
Miniature Lupine	<i>Lupinus bicolor</i>	2026-02-18
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2023-09-02
Narrowleaf Willow	<i>Salix exigua</i>	2026-07-07
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2024-04-27
pale smartweed	<i>Persicaria lapathifolia</i>	2026-05-06
palmate salty bird's-beak	<i>Chloropyron palmatum</i>	2025-05-28
Purslane Speedwell	<i>Veronica peregrina</i>	2024-04-12
Red Ribbons	<i>Clarkia concinna</i>	2026-05-05
Red Ribbons	<i>Clarkia concinna concinna</i>	2023-06-14
redwhisker clammyweed	<i>Polanisia dodecandra</i>	2026-06-18
rough cocklebur	<i>Xanthium strumarium</i>	2023-09-02
Sacred Datura	<i>Datura wrightii</i>	2025-10-06
Saltgrass	<i>Distichlis spicata</i>	2025-07-03
sea clubrush	<i>Bolboschoenus maritimus</i>	2024-05-15
shield-bracted monkeyflower	<i>Erythranthe glaucescens</i>	2026-05-07
showy milkweed	<i>Asclepias speciosa</i>	2023-07-28
shy monkeyflower	<i>Erythranthe nasuta</i>	2025-04-11
Silvery Bryum	<i>Bryum argenteum</i>	2025-12-27

COMMON NAME	SCIENTIFIC NAME	LAST OBS
swamp smartweed	<i>Persicaria hydropiperoides</i>	2026-03-02
tall flatsedge	<i>Cyperus eragrostis</i>	2026-06-18
tarweed fiddleneck	<i>Amsinckia lycopsoides</i>	2024-04-12
Telegraphweed	<i>Heterotheca grandiflora</i>	2026-07-07
Toyon	<i>Heteromeles arbutifolia</i>	2026-01-22
turkey mullein	<i>Croton setiger</i>	2023-06-28
turkey tangle frogfruit	<i>Phyla nodiflora</i>	2025-07-04
twining snakelily	<i>Dichelostemma volubile</i>	2024-05-11
Upright Pepperweed	<i>Lepidium strictum</i>	2024-04-12
valley oak	<i>Quercus lobata</i>	2026-06-13
water fern	<i>Azolla filiculoides</i>	2025-07-03
Whiskerbrush	<i>Leptosiphon ciliatus</i>	2025-04-10
Wild Licorice	<i>Glycyrrhiza lepidota</i>	2024-06-06
woolly rosemallow	<i>Hibiscus lasiocarpus</i>	2025-07-15

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 39.2147, -122.0140 This site supports a **riparian-alkali transition community** dominated by cottonwood, box elder, and tule, with abundant alkali-tolerant species (alkali heliotrope, alkali mallow, big saltbush) indicating saline or alkaline soil conditions. The ecology reflects a **semi-arid floodplain or drainage corridor** with seasonal water availability, where native shrubs and herbaceous plants are adapted to fluctuating moisture and elevated salts. Grading and drainage activities must account for **poorly drained to seasonally saturated soils** in lower areas and potential salt accumulation in surface layers; any site work should preserve riparian connectivity and avoid disrupting seasonal water flow patterns that sustain this plant community. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

25	30	9	82
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants

VENOMOUS SNAKES

SPECIES	OBSERVATIONS
<i>Crotalus oreganus oreganus</i>	25 obs
<i>Crotalus oreganus</i>	25 obs

VENOMOUS SPIDERS

SPECIES	OBSERVATIONS
<i>Latrodectus hesperus</i>	30 obs

STINGING INSECTS

SPECIES	OBSERVATIONS
<i>Dasymutilla aureola</i>	9 obs
<i>Dasymutilla sackenii</i>	9 obs

TOXIC PLANTS

SPECIES	OBSERVATIONS
<i>Toxicodendron diversilobum</i>	39 obs
<i>Datura stramonium</i>	21 obs
<i>Datura ferox</i>	21 obs
<i>Datura wrightii</i>	21 obs
<i>Conium maculatum</i>	19 obs
<i>Urtica gracilis</i>	3 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Briefing – Lake County Area Your highest-priority hazards are **Northern Pacific rattlesnakes** (common in this region) and **poison oak**, which dominate the landscape. For snakes: wear closed-toe boots, watch where you place hands/feet, and make noise while moving—they'll typically retreat. For poison oak and other toxic plants: learn to identify three-leaflet clusters, wear long sleeves/pants, wash clothing immediately after fieldwork, and never burn vegetation. **Widow spiders** hide in dark spaces and under objects, so shake out equipment and inspect before handling. Peak snake and stinging insect activity runs April through October, when temperatures rise and plants are dense—be especially vigilant during warm months and early morning/dusk surveys. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FAA AIRWAYS — ATS Routes crossing site area

22
Total Airways

8
Jet Routes

6
Victor Airways

8
Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	342.8°	161.8°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J189	Jet Route	High	18,000 ft	—	335.5°	155.0°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	22,000 ft	—	342.6°	162.6°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	162.7°	343.3°
T259	T-Route	Low	—	—	340.5°	160.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°
V150	Victor Airway	Low	3,000 ft	—	35.4°	215.5°
V195	Victor Airway	Low	5,300 ft	—	352.9°	172.9°
V199	Victor Airway	Low	9,000 ft	—	21.3°	199.6°
V200	Victor Airway	Low	6,200 ft	—	72.5°	252.8°
V23	Victor Airway	Low	2,000 ft	—	293.8°	113.6°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

3
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Mainline	1	No	No	WEST VALLEY
CFNR	Other	1	No	No	—
CFNR	S	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200
Total Observations

88
Total Species

70
Birds

8
Mammals

8
Reptiles

2
Amphibians

BIRDS (70 species)

COMMON NAME	Scientific Name	OBS
Great Egret	<i>Ardea alba</i>	8
Black-crowned Night Heron	<i>Nycticorax nycticorax</i>	8
Western Cattle-Egret	<i>Ardea ibis</i>	7
Red-winged Blackbird	<i>Agelaius phoeniceus</i>	6
Killdeer	<i>Charadrius vociferus</i>	5
Bald Eagle	<i>Haliaeetus leucocephalus</i>	5
Mourning Dove	<i>Zenaida macroura</i>	4
Wild Turkey	<i>Meleagris gallopavo</i>	4
House Sparrow	<i>Passer domesticus</i>	4
Bullock's Oriole	<i>Icterus bullockii</i>	4
House Finch	<i>Haemorhous mexicanus</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	3
Western Kingbird	<i>Tyrannus verticalis</i>	3

COMMON NAME	Scientific Name	OBS
Pied-billed Grebe	Podilymbus podiceps	3
American Bittern	Botaurus lentiginosus	3
+ 55 additional species		
MAMMALS (8 species)		
COMMON NAME	Scientific Name	OBS
Columbian Black-tailed Deer	Odocoileus hemionus columbianus	4
Black-tailed Jackrabbit	Lepus californicus	4
North American River Otter	Lontra canadensis	3
Mule Deer	Odocoileus hemionus	3
Coyote	Canis latrans	1
Pronghorn	Antilocapra americana	1
American Badger	Taxidea taxus	1
Domestic Cattle	Bos taurus	1
REPTILES (8 species)		
COMMON NAME	Scientific Name	OBS
Pacific Gopher Snake	Pituophis catenifer catenifer	5
Western Fence Lizard	Sceloporus occidentalis	3
Mountain Garter Snake	Thamnophis elegans elegans	3
Gopher Snake	Pituophis catenifer	2
Northwestern Pond Turtle	Actinemys marmorata	2
Western Yellow-bellied Racer	Coluber constrictor mormon	2
Red-eared Slider	Trachemys scripta elegans	1
Northern Pacific Rattlesnake	Crotalus oreganus oreganus	1
AMPHIBIANS (2 species)		
COMMON NAME	Scientific Name	OBS
Western Toad	Anaxyrus boreas	2
Western Spadefoot	Spea hammondi	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in northern California supports a diverse wildlife community characteristic of mixed grassland and riparian habitats with proximity to wetland or lacustrine features, as evidenced by the prominence of wading birds (egrets, herons), waterfowl-associated species, and aquatic reptiles (pond turtle, red-eared slider). The presence of regulatory-concern species including Bald Eagle (BGEPA), pronghorn (state-sensitive), and sensitive amphibians (Western Spadefoot and Western Toad) indicates intact native habitat with limited development; the diverse herpetofauna and predator guild (coyote, badger, river otter) further suggest relatively low fragmentation and good connectivity. However, the occurrence of species tolerant of anthropogenic disturbance (domestic cattle, house sparrow, red-eared slider) and wading birds attracted to agricultural drainage suggests the landscape is actively managed for livestock and agriculture, warranting careful evaluation of potential project impacts on seasonal water availability and wildlife corridors. Any proposed development should incorporate wetland/riparian setbacks, avoid habitat conversion during sensitive breeding seasons (particularly for eagles and ground-nesting birds), and maintain landscape permeability for mammalian movement. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 39.21°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = 39.2° (latitude) → 10/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	96.9%	good
5/12	22.6°	97.5%	good
6/12	26.6°	98.1%	↔ excellent
7/12	30.3°	98.7%	↔ excellent
8/12	33.7°	99.2%	↔ excellent
9/12	36.9°	99.7%	↔ excellent
10/12	39.8°	99.9%	↔ annual optimal
12/12	45°	99.1%	↔ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	49	30	Minimize south glass
South glass (SHGC 0.6)	73	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	26	104	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	233	301	Highest load surface
Roof — light (absorb 0.35)	86	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	61	79	Best roof option for heat
South wall — dark	91	69	
South wall — light	30	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3B Hot-Dry: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.2	2	2.4	OK
30ft	All year	3.1	2.9	3.8	Zone 2 check
40ft	All year	4.1	3.8	5.4	△ verify
50ft	All year	4.9	4.7	7.8	△ verify
60ft	All year	5.7	5.5	9.2	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†Balanced = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🏠 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan Your location gets fantastic sun—about 6.6 peak hours daily on average, jumping to 9 hours in summer and dropping to 3.9 in winter, which means solar panels here will work harder than in most places and pay for themselves faster. The roof is critical: a dark roof acts like a heat-absorbing sponge in summer, pulling in 301 BTU per hour per square foot (think of it as thousands of tiny heaters mounted on your home), but switching to a light color cuts that nearly in half and saves your air conditioning from working overtime. Your roof's optimal angle is about 39 degrees, which conveniently matches a 10/12 pitch—something worth checking with your architect since it's the sweet spot for both winter and summer performance. Now here's something most people miss: your west-facing windows are your afternoon weak spot, soaking up 104 BTU per hour in summer heat, so shading or reflective glass there makes a huge difference, while south-facing glass is actually helpful in winter for free warmth. A mature deciduous tree on the south side is genuinely smart—it'll shade your house about 4 hours in summer (cooling it down) and drop its leaves in winter to let warming sun through; the only caution is if you're in a wildfire zone, where trees closer to the home create risk, so check your local WUI (wildland-urban interface) guidelines first. Thermal mass—think of it as your home's built-in battery for temperature—means heavy materials like concrete or adobe soak up heat slowly and release it slowly, so a thick-walled home holds steady temperatures longer than a lighter-framed one, which is especially valuable in your hot-dry climate. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

Site Intelligence Briefing 39.21470°N, 122.01400°W | Colusa County, CA

Structural Design Drivers & System Selection

This site falls within IECC Climate Zone 3B (Hot-Dry), characterized by an annual freezing index of 9°F-days. Wind, snow, and seismic parameters are not reported for this location, meaning the design team will rely on default IECC provisions or historical records for lateral loading. The absence of reported seismic data does not indicate zero seismic hazard; California Building Code requires seismic design in all zones. Without site-specific wind or snow loads reported, thermal cycling and diurnal temperature swings in a hot-dry climate become the dominant environmental stressor, favoring low-slope roof systems and controlling expansion joints. The structural frame should be selected on the basis of soil-bearing capacity and seismic demand per IBC rather than wind or snow governance.

Geotechnical Foundation Requirements

The site is underlain by Moonbend silt loam with slopes of 0 to 2 percent, classified as Hydrologic Group C with moderately well-drained conditions. Group C soils (silt and clay loams, silty clay) typically exhibit moderate permeability and moderate shrink-swell potential; CBR values are commonly in the 5–8 range for this soil class, supporting light to moderate foundation loads via shallow footings. The frost depth of 6 inches, derived from Brown's Formula using a 9°F-day AFI, governs footing depth; local code will likely require frost protection below 6 inches or frost-protected shallow foundations (FPSF) per IECC 403.3 where applicable. Soil texture is incomplete in the dataset; recommend on-site boring to confirm clay fraction, consolidation behavior, and confirm drainage classification under saturated conditions. Drainage design should route surface and subsurface water away from foundations given the moderately well-drained classification, which implies some seasonal saturation risk in lower terrain.

Hazard Assessment

No Quaternary faults are mapped within the search radius, and no subsidence basin data is associated with this location. However, this absence does not eliminate seismic risk; the site is in Colusa County, California, and seismic design per IBC Chapter 11 and California Building Code Chapter 11B is mandatory. Snow load data is not reported, consistent with the low AFI and hot-dry climate designation. Wildfire hazard assessment is not provided in the dataset; however, vegetation and fire risk should be verified against CalFire's Fire Hazard Severity Zone maps for the specific site. No evidence of active subsidence or faulting reduces long-term foundation distress risk compared to areas with recorded tectonic or hydrogeologic instability.

Site Context: Mining & Mineral Resources

One active mineral producer, Cortina Mill (SDG commodity, likely sand/gravel/aggregate), operates 0.6 miles from the site. The MRDS database records approximately 200 mineral deposits within a 35-mile radius, typical of California's resource-rich geology; none appear to present direct subsurface or blast hazard at this location. Verify zoning and operational permits for the Cortina Mill to confirm distance estimates and assess dust, noise, or vibration impacts on structure-sensitive equipment or habitability. No active oil and gas wells currently produce within 1.2 miles; the nearest TXO Production Corp. well is plugged, eliminating subsidence and pressure-decline risks associated with hydrocarbon extraction. Historical well activity should be cross-referenced with county records to confirm casing integrity and ground stability.

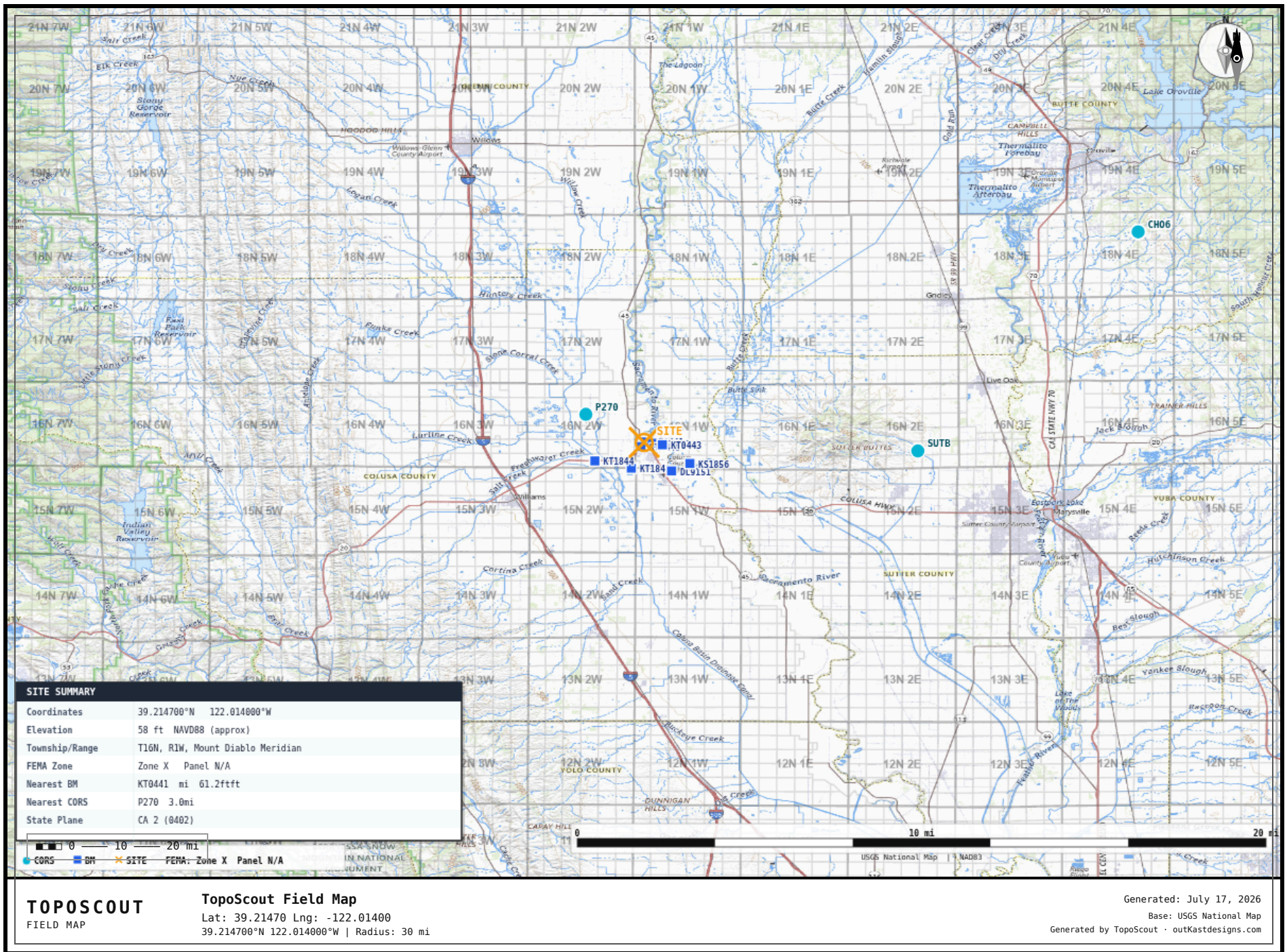
Bottom Line

Initiate foundation design with a geotechnical site investigation (auger boring, laboratory Atterberg limits, and bearing-capacity tests) to refine soil drainage and CBR, confirm 6-inch frost depth compliance, and validate footing type selection against Moonbend silt loam behavior under design loads.

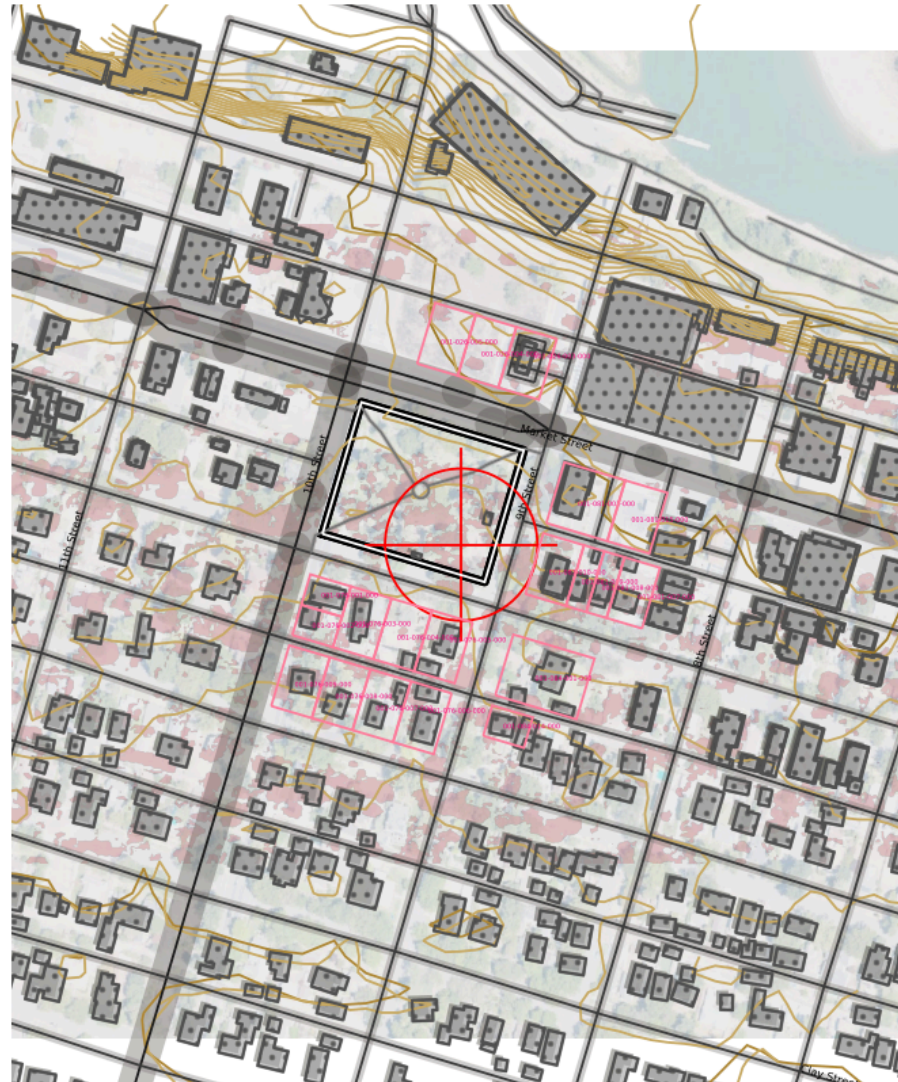


Copy to clipboard

Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

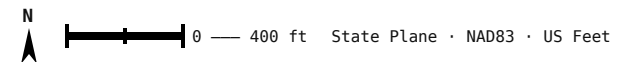


TOPOSCOUT SITE PLAN PREVIEW



INCLUDED LAYERS

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QO
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polylines (all geometry)





SITE AERIAL — 600 ft × 600 ft

39.21470°N, 122.01400°W · Google Aerial



AREA CONTEXT — 3000 ft × 3000 ft

39.21470°N, 122.01400°W · Google Aerial

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

